

Magma & LMFDB Mini-Course

Project A: Contributing to the LMFDB Lattice Database (alpha)

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Why this project

The LMFDB lattice database is being substantially redesigned for the next major release. The current *production* (beta) version at <https://www.lmfdb.org/Lattice/> is **obsolete** and being replaced; the active development happens on the **alpha** version at <https://alpha.lmfdb.org/Lattice/>, with a new schema involving two main tables:

- **lat_lattices_new**: one entry per integral lattice up to isomorphism, with a rich invariant package including multiple discriminant variants, density, kissing number, shortest vectors, level, automorphism group, etc.
- **lat_genera**: one entry per genus, addressing the long-standing request that genera have their own pages (issue #748 from 2016).

This redesign has coverage gaps you can help fill. As you may have noticed during the scavenger hunts:

- The lattice E_{16} (a.k.a. D_{16}^+) – the second even unimodular lattice of rank 16 from Lecture 1 – is in the alpha database, but its **root sublattice** is not yet computed.
- The 24 **Niemeier lattices** (even unimodular rank 24, including the Leech) are not yet present in either table in a fully searchable form, even though they are central examples in the theory.
- Many **small ternary and quaternary genera** from Brandt–Intrau–Schiemann and Nipp’s tables can be cross-checked against the new **lat_genera** schema, and some may be incomplete or missing their links to constituent lattices.

This project: **help close those gaps in the alpha version**. Even a single new genus with carefully-computed invariants in the new schema is a real contribution; a coherent batch (e.g., all 24 Niemeier lattices, or all genus pages of small ternaries with class number ≥ 2) could lead to a publishable contribution.

The new schema in brief

The schemas are documented in detail at https://github.com/roed314/k3s-lmfdb/blob/main/schemas/lat_lattices.md and https://github.com/roed314/k3s-lmfdb/blob/main/schemas/lat_genera.md, with live field-level documentation at <https://alpha.lmfdb.org/api/>. The high-level structure:

Label. Each lattice has label of the form $r.s.d.j_1.j_2....j_k.x$, where:

- r is the rank;
- $s = n_+$ is the number of +1s in the diagonalization over $\mathbb{R}$ (so a positive-definite rank- n lattice has $s = n$);
- d is $|\det G|$ (the absolute determinant of the Gram matrix);
- $j_1, \dots, j_k$ encode the Jordan decompositions of the local genus symbols at primes p with $p^2 \mid 2d$, in base 62;
- x is a hexadecimal string encoding the local types/oddities/signs of the Jordan blocks;
- For definite lattices, the final “tiebreaker” uses lexicographic ordering on canonical Gram matrices.

Per-lattice invariants (in `lat_lattices_new`): `rank`, `nplus`, `nminus`, `disc_abs`, `disc_sign`, `disc_radical`, `disc_witt`, `disc_geometric`, `disc_quadratic`, `disc_half`, `disc_2adic_unit`, `bad_primes`, `class_number`, `density`, `center_density`, `covering_norm`, `hermite`, `kissing`, `shortest`, `level`, `minimum`, `aut_size`, `aut_label`, `aut_group`, theta series coefficients, root system data, and several others. The schema lists each column with its type and a link to its definition in the LMFDB knowledge base.

Per-genus invariants (in `lat_genera`): `genus symbol`, `rank`, `nplus`, `nminus`, `disc_abs`, `disc_sign`, `disc_radical`, `disc_witt`, `disc_geometric`, `disc_quadratic`, `disc_half`, `disc_2adic_unit`, `bad_primes`, `mass` (Smith–Minkowski–Siegel mass formula), `class number` (number of classes in the genus), and several others. The schema lists each column with its type and a link to its definition in the LMFDB knowledge base.

Three phases

Phase 1 – One lattice end-to-end (4–6 hours).

A great starting point: choose a single Niemeier lattice (say $N(D_{24})$ or $N(A_{17}E_7)$, the first two entries in the SPLAG list) and compute the full invariant package for the new alpha schema.

1. Construct the lattice in Magma. Conway–Sloane *Sphere Packings, Lattices and Groups* (SPLAG), Chapter 16, Table 16.1, gives an explicit “glue construction” for each Niemeier lattice from its root sublattice and a glue code. Magma’s `Lattice("A", n)`, `Lattice("D", n)`, `Lattice("E", n)` produce the root lattices; the glue vectors are listed in SPLAG.
2. Compute the standard invariants: `Determinant(GramMatrix(L))`, `Minimum(L)`, `KissingNumber(L)`, `#AutomorphismGroup(L)`, `ThetaSeries(L, 30)`.
3. Compute the *enriched* invariants that the new schema requires beyond what the legacy schema asked for: Witt discriminant, the various density measures (centered, geometric, quadratic, half), Hermite number, level. Definitions are in the LMFDB knowledge entries linked from the schema file.
4. Identify the root sublattice:
`R := sub<L | [v : v in MinimalVectors(L) | InnerProduct(v, v) eq 2]>`, then `Index(L, R)` for the gluing index.
5. Compute the label `r.s.d.j_1....x` following the algorithm in the schema documentation. (For a positive-definite even unimodular lattice of rank 24, this label has the form `24.24.1.0.tiebreaker`; the `genus_spec` digit is 0 because the signature is a multiple of 8 and the lattice is even.)

6. Format the data for postgres per the schema and submit it to me.

Tip. Start with $N(D_{24})$: a single root component, simplest glue code, well-known invariants from SPLAG Table 16.1.

Phase 2 – A coherent family (10–20 hours).

Once your pipeline produces clean data for one lattice, scale up to a batch. Some good choices:

- **All 24 Niemeier lattices.** Iterate through Table 16.1 of SPLAG; the bottleneck is automorphism group orders for the large root systems ($24A_1$ involves the Mathieu group M_{24} and requires care). Each lattice becomes a `lat_lattices_new` entry; collectively they all lie in the same genus (rank 24, signature 24, $|\det| = 1$, even), so they constitute one `lat_genera` entry with 24 constituent lattices.
- **Both even unimodular lattices of rank 16** ($E_8 \oplus E_8$ and E_{16}). Compute root sublattices for both (currently missing for E_{16}); these constitute the unique genus of rank-16 even unimodular lattices, so the `lat_genera` entry should have exactly two constituents and reflect the Witt identity $\theta_{E_8 \oplus E_8} = \theta_{E_{16}}$.
- **The genus of Λ_1** (the lecture’s running rank-3 lattice, $|\det| = 22$). The genus has class number 2; the `lat_genera` entry should link to both class representatives. The unique-spinor-genus fact (which we used in Hunt 1) belongs in the genus page.
- **Rank-3 ternary genera with $|\det| \leq 100$ and class number ≥ 2 .** The Brandt–Intrau–Schiemann tables at Nebe’s online catalogue (<https://www.math.rwth-aachen.de/~Gabriele.Nebe/LATTICES/>) give the complete list of forms; many of these genera are present in the alpha database but with incomplete data. Audit and complete.

Phase 3 – Knowledge entries and cross-references (after CTNT).

After your data is staged for ingestion, return to write or polish the “knowledge entries” – the Wikipedia-style descriptions for each object or concept. Many lattice entries are marked *awaiting review* (e.g., the entry for the D_4 lattice at <https://www.lmfdb.org/knowledge/show/lattice.4.4.2.1.1.top>).

Particularly valuable cross-references (these turn isolated lattice pages into nodes in the LMFDB web of mathematics):

- Each *even unimodular* lattice’s theta series is an element of $M_{r/2}(\mathrm{SL}_2(\mathbb{Z}))$ where r is the rank. For the Niemeier lattices, this is a weight-12 form, lying in the 2-dim space spanned by E_4^3 and Δ (cf. Hunt 4.4). Link each Niemeier lattice page to the explicit expression $\theta_\Lambda = aE_4^3 + b\Delta$ and to the modular forms database entry – which itself requires the Eisenstein series E_4^3 to be properly indexed in LMFDB, an open issue Project B addresses.
- The genus of Λ_1 (rank 3, $|\det| = 22$) carries a half-integral weight $3/2$ theta-difference; its Shimura lift is the elliptic newform **11.2.a.a** (Hunt 1 bonus). Document this connection in the genus page; this is the kind of cross-domain link the LMFDB is uniquely positioned to host.
- Genera should link to their *spinor genus* structure, mass formula value, and (where known) to the corresponding cusp forms via Eichler’s correspondence. The `lat_genera` schema is being designed with these links in mind.

Resources

- Alpha LMFDB lattice database: <https://alpha.lmfdb.org/Lattice/>.
- Alpha LMFDB API (field-level documentation): <https://alpha.lmfdb.org/api/>.
- Schema documentation:
 - https://github.com/roed314/k3s-lmfdb/blob/main/schemas/lat_lattices.md
 - https://github.com/roed314/k3s-lmfdb/blob/main/schemas/lat_genera.md
- Conway & Sloane, *Sphere Packings, Lattices and Groups* (SPLAG), 3rd ed., Springer, 1999. Chapter 16 for Niemeier lattices.
- Nebe’s online catalogue: <https://www.math.rwth-aachen.de/~Gabriele.Nebe/LATTICES/>.
- Magma: Lattice, ThetaSeries, AutomorphismGroup, GenusRepresentatives, Neighbours.
- Sage: IntegralLattice, the `sage.modules.free_quadratic_module_integer_symmetric` module.

Where to ask for help. The LMFDB lattice editors include Eran Assaf, David Roe, and John Voight; David Roe maintains the `k3s-lmfdb` schema repository. Issues can be filed at <https://github.com/LMFDB/lmfdb/issues>.